Date :
Page :

5/09/2024

Engineering materials

* Classification of Engineering materials

- Metals (naturally available in earth)
  - Ferrous
    - iron is the main constituent
    - Carbon
    - Eg:- Steel, Wrought Iron, cast iron.
    - good magnetic property
    - ductile
  - Non-ferrous
    - iron is not present
    - no magnetic property
    - no weight
    - Eg:- Al, Cu (तांबा), Bronze (कांस्य / कासा), (पीतल) Brass, Zn, Pb, Ni, Au, Ag, Ti, etc.
- Non-metals
- Organic polymers
- Composite materials

* 6 Workshops

- welding shop
- machine shop (lathe machine) (खराद)
- fitting shop
- Carpentry shop
- Blacksmith shop
- forging shop

2 practice in each shop so total 12 workshops

task of engineering material is to bear the load.

deflection - change of shape

* Engineering materials refer to a group of materials used in construction of man-made structures and mechanical components.
* Materials must withstand (able) applied loading without breaking and excessive

deflection.

deflection

- Elastic (regain its shape after removing the load)
- Plastic (deformation or elongation in the material)

- The selection of materials depends on the factors such as mechanical, thermal and electrical properties, cost, availability and environmental factors.

* Metals

Metals are capable of changing their shape permanently, they have good thermal (heat) and electrical conductivity. Metals may be ferrous or non-ferrous, the behaviour and properties of ferrous metals depends upon the percentage and the form of iron and carbon present in them.

Only 0 to 6.67 % of carbon can be diffused in Fe.

Fe-C diagram (Not in course)

* Alloy

- An alloy is prepared by melting together two or more metals.
- They possess properties quite different from

those of their consituent metals.

steel + Cu → corrosion resistance

- Alloys may be ferrous or non-ferrous depending on the base metal used.
- Metals and alloys have found maximum use among other engineering materials because of their useful properties.
- Eg:- Strength, hardness, ductility, malleability formability, weldability (ease of welding), machinability (material removal process by adding Phosphate in metal), electrical and thermal conductivity.

* Non-metals
- Ceramics and glasses are non-metallic inorganic substances, they are compounds of metallic and non-metallic elements. They are brittle and have good insulating properties against heat and electricity, i.e; bad conductor of heat and electricity.
- Eg:- rocks, glass, minerals, fibre glass, abresives, cement, silica, free clay.

* Organic polymer
- They are usually consisting of carbon chemically combined with hydrogen, oxygen, nitrogen & other non-metallic substances.
- Polymers are obtained from the monomers or chemically combined to form a long chain. They are relatively light in weight which,

Alloy - only metals
Composite - any atom like metal, non-metal, Organic polymers, etc.

they have high degree of plasticity.

- Eg:- polyethylene, terylene, nylon, cellulose, cotton, natural & synthetic rubber, teflon and a Bakelite, etc.

* Composite materials

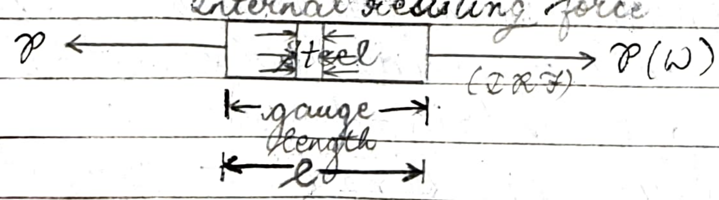
- Composites are made up of two or more materials of dissimiliar properties. They may be combining as metals & ceramics or metals polymers, ceramic polymers or other combination.
- Eg:- Reinforced materials (main constituent of CM) (force is induced by increasing its strength)
One of the constituents is called as reinforced materials. Eg:- Steel reinforced concrete (non-metal i.e., ceramics), vinyl coated steel (Alloy + polymer), fibre reinforced plastic (ceramic) (polymers) and carbon reinforced rubber

Imp * The Properties of engineering materials

- Engineering materials possess a wide range of properties. An engineer must possess a thorough knowledge of these properties in order to put them into practical use.

Imp → Mechanical Properties

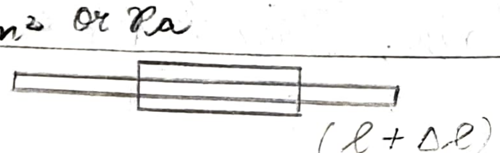
load or external force
internal resisting force
÷ by A_c



UTM

$$Stress = \frac{IRF}{A_c} = N/m^2 \text{ or } Pa$$

$$Strain = \frac{dl}{l}$$



- Those properties of materials that describe their behaviour under external mechanical forces or external load.
- These include strength, stiffness, elasticity, ductility, plasticity, toughness, brittleness, hardness, malleability and creep.

1.) Strength :- The strength of material is its ability to resist externally applied force without failure. The stronger material offers higher resistance to deformation.

2.) Stress: The internal resistance offered by a part per unit area to an externally applied force is called stress. (N/m^2). Denoted by (σ)

3.) Elasticity - It is a property of material, due to which it by which it regains comes back to its original shape after deformation when the external load is removed.

or

Elasticity is the ability of a material to resist permanent deformation under a load.

→ Hooke's law

$\sigma \propto \epsilon$ upto elastic limit

$\boxed{\sigma = E\epsilon} \rightarrow$ Strain $\epsilon \rightarrow$ Strain

$E \rightarrow$ Young's modulus of elasticity

E strength = 200 lakh ⟶ σ permissible

Factor of safety = $\dfrac{\text{Strength of material}}{\sigma_{\text{permissible}}}$

$2 = \dfrac{200}{\sigma_{\text{permissible}}} \Rightarrow \sigma_{\text{permissible}} = 100$ MPa stress ∝ strain upto P

Date :
Page :

yielding stress phenomenon

4.) Plasticity :- It is the ability of a material to retain the deformation permanently. Due to this property, various metals can be converted into desired shape.

5.) Ductility :- It is the ability of a material to be elongated or drawn into thin wires without breaking. This property ~~to~~ relates to tensile load. Metals in order of decreasing ductility are :-
Steel, Cu, Al, Ni, Zn, ~~Sn~~, Pb

gold & silver more ductile

6.) Brittleness :- It is the property that resists elongation or distortion generally beyond elastic limit. The brittle material breaks under tension with neglible or little plastic deformation.
Eg :- Gray cast iron, glass, ceramics, Graphite, concrete, etc.

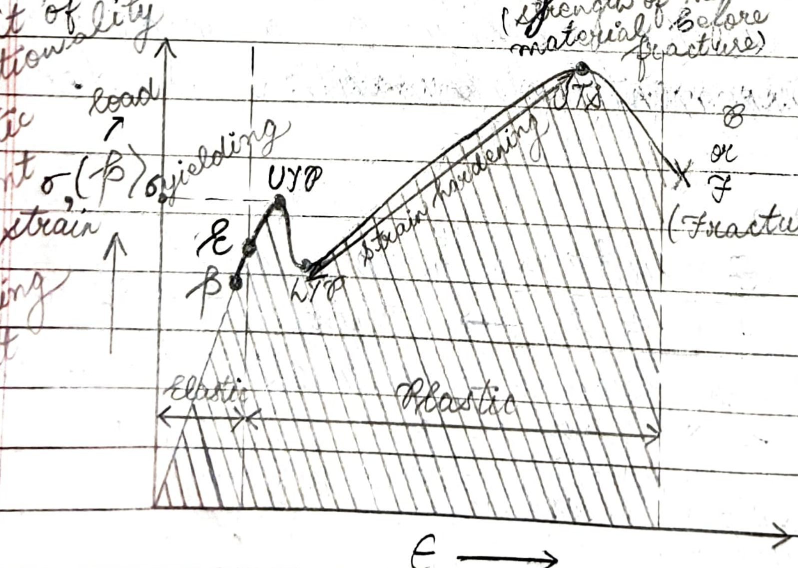
elasticity retain its original shape

(load)

* Stress - Strain diagram

(low carbon steel)
Mild Steel
(Ductile material)
(Lowest carbon %)

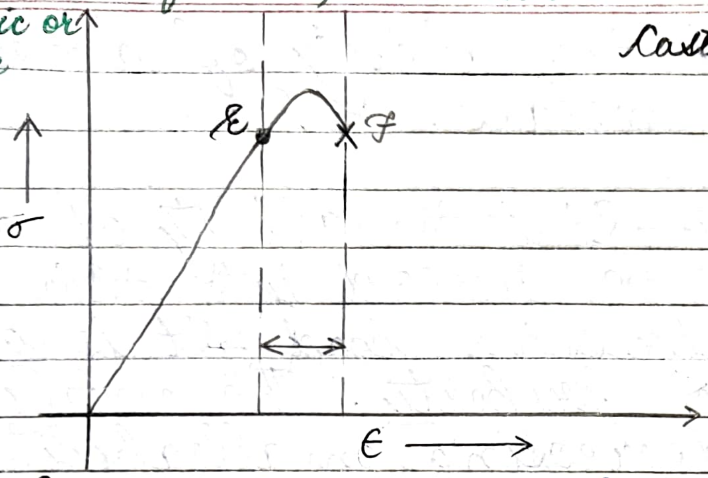
$\sigma \propto \epsilon$
P → limit of proportionality
E - elastic point
B - Breaking point



(plastic deformation) yielding
UYP - upper yield point
UTS - Ultimate Tensile strength
LYP - lower yield point
The less stresses are required for plastic deformation

Indentation hardness tests are used in mechanical engineering to determine the hardness of a material to deformation. Several such tests exist, wherein the examined material is indented until an impression is formed; these tests can be performed on a macroscopic or microscopic scale.

Cast iron



glass - hard not tough

Impact load also known as sudden load

7.) Toughness :- It is the ability of a material to resist fracture due to impact load. Toughness indicates the amount of energy absorbed by the material in the both elastic or plastic region upto fracture. (or deformation)

diamond hardest material expensive

Toughness = area under the curve give us the amount of energy.

8.) Resilience :- It is the property / Resilience is the ability of a material to absorb the energy within elastic limit. This stored energy is released when applied load is removed.

scratch, identation, deformation

9.) Hardness :- It is the ability of a material to resist, bear scratch, deformation & indentation. Hardness gives a general indication of strength of material. It is also defined as resistance to permanent identation.

hard - graphite, tungsten, molybdenum, knife used to cut soft material

HSS - High speed strength

To measure hardness we perform
- Vicker hardness test
- Brinell hardness test
- Rockwell hardness test

Steel is more elastic than rubber

Hardness is the ability of a metal to cut another metals.

9) Malleability - It is the ability of a material to be pressed into thin sheets under compressive load. It is a compressive property. The metals in order of decreasing malleability are:- Lead, steel, Wrought Iron, Cu, Al, Ag, Au, etc. (ductile & malleable)

10) Creep - When a part is subjected to a constant static load at elevated temperatures (increased temperatures) for a period of time, it undergoes permanent deformation at slow rate called creep.
Eg:- Steam turbines, gas turbines, Rockets turbine, missiles turbine.
Eg:- Molybednum (Mo)

* Electrical Properties
The ability of a material to allow or resist the flow of current.
Eg:- Conductivity, resistivity, thermoelectric strength, dielectric strength

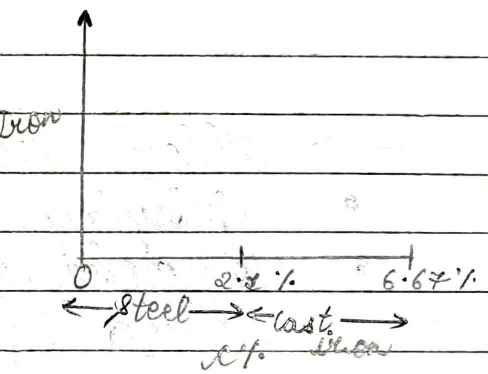
* Chemical Properties
Reactions with materials when they come in contact with hostile शत्रुतापूर्ण environment प्रतिरोधी (resistive) like chemicals, environmental pollution, air, water, etc. destructive

Eg:- Corrosion, oxidation, degradation, rusting.

* Thermal Properties
Thermal Properties are when the materials to be exposed to varying temperatures.
Eg:- Thermal conductivity (ability of a material to conduct heat), specific heat, melting point, freezing point & boiling point.

17/10/24

Iron + Carbon ≈ 6.67% Steel



- Plain carbon steel (3 types)
- Alloy steel
- Stainless steel
- tool steel

* Steel is produced from the given furnace method:-
1. Crucible furnace
2. Open hearth furnace
3. Electric furnace to produce the steel from iron
4. Bessemer furnace

pig-iron blast furnace
cast-iron cupola furnace

Date :
Page :

→ Plain carbon steel
Based on the % of carbon softness, ductility
- cost ↑ ductility ↓ low carbon steel (LCS) or mild steel (MS) low cost
  C% → upto 0.25 %
- C% ↑ Medium carbon steel (MCS)
  C% → 0.25 − 0.6 %
- High carbon steel
  C% → 0.6 − 2.1 %
  0.6 − 1.5 % (for engineering application)

Increases ductility decrease, lesser the C% cost will increase

→ Alloy steel
- Alloying elements (Any ferrous or non-ferrous metal)
- To achieve the desired properties like weldability, machinability, ductility, creep, hardness, toughness, stiffness, etc.
- alloyic properties of steel (Mn, Si, Ph, S, Cu, Ni, Mo, W, Cr, etc.)

lustorous properties (SS) expensive
→ Stainless steel (no rusting) used in utensils
- Major element / constituent is Chromium (Cr)
- Minimum 10.5% Cr is present.

Passive layer of (Cr_2O_3) — This gives lustorous property & prevents rusting

- Types of stainless steel
1. Ferritic S.S
2. Martintic S.S Martensitic S.S
3. Austenitic S.S
4. Duplex S.S

Date :
Page :

5) P.H. S.S
(Precipitation hardening)

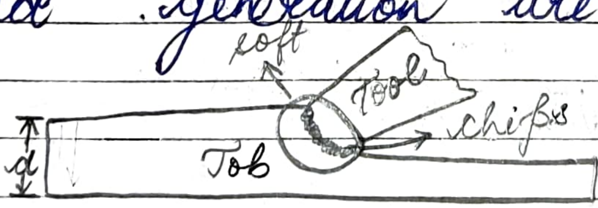
→ Tool steel → must be hard → cutting tool
→ Dies / Dyes

- Tool steel is made up of High carbon steel (H.C.S)
- % C → 0.6 – 1.5% (low speed में use करते हैं)
- High speed steel (H.S.S)

→ High speed steel (H.S.S) H.C.S can only perform for small & low speed jobs

Machine Shop

This steel is used for making cutting tools which are used to cut the metal at higher cutting speed. When the tool is made of high carbon steel (H.C.S), It cannot retain their sharp cutting edge under heavy loads at higher cutting speed. This is due to the fact that a high cutting speed sufficient creep is generated at high temp. of cutting zone which make the tool material soft & the tool is not able to cut efficiently for longer period. The factors responsible for high speed generation are –



1) Higher cutting speed
2) Heavy cut
3) Hardness of the material being machined.
4) High friction (घर्षण) b/w the tool & job interface.

Date :
Page :

- However, high speed steel (H.S.S) has the valuable property of retaining their hardness even when heated to red heat. This property of a material to retains its hardness at elevated temperatures is known as red hardness.
- The H.S.S has high value of red hardness. Most of the H.S.S contain Tungsten (W) as chief alloying elements but other elements like Co, Cr, Venedium, etc. may be added in some proportion.
- The most commonly used H.S.S is 18-4-1 Steel which contains :-

  18% → W (Tungsten)
  4% → Cr (Chromium) rest is iron
  1% → V (Venedium)
  C% is 0.8 - 1.2 %.
- Eg:- Single point cutting tool used in lathe machine, shaper and cleaner machine tools, milling cutters, drill, ~~tr~~ broaches, slotter's tool.

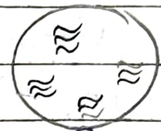
ढलवा लोहा Casting - Foundry shop

* Cast iron (C.I)
→ M.P - 1250 °C
→ Nature - Brittle material
→ They are weaker in Tension
→ Stronger in Compression
→ Forging (Blacksmithy) cannot be occur.
→ Malleability is less.

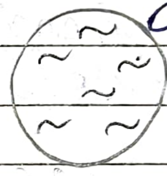
→ It is made from pig iron.
→ Formation

Pig iron $\xrightarrow[\text{furnace}]{\text{cupola}}$ C.I.
(Coke & lime)

→ Low cost
→ Good castability



→ Carbon can exist in iron in two forms:-
1.) • Combined form or Fe_3C or cementite
• Cast iron will be hard comparatively.
2.) • Free form (Graphite flakes)
• Self lubricating property
• & Vibration damping



→ Types of cast iron
1.) Grey C.I.
2.) White C.I.
3.) Chilled C.I.
4.) Spheroidal or Nodular or ductile C.I.
5.) Malleable C.I.

formation:- melt then solidify

(hard C.I.) %C → (2 – 4%)
1.) ~~Grey~~ White C.I. :- property in hardness
• High wear and Abrasion resistant
removal of the material from surface due to striking
• It is hard and brittle.
inner soft outer hard
• Fast cooling — Inter-connected
therefore
• Weak in Si content (< 1%)
• Presence of C form (Combined form) (Fe_3C)

(cooling rate)
Solidification Rate
fast slow

Si content कम तो cooling जल्दी होगी और ~~क्योंकि~~ जल्दी cooling होगी इसका मतलब carbon combined form में present है

Date :
Page :

* Application (spiral shape) made of brass (alloy of Cu)

water has pressure heat of nozzles converting it into velocity

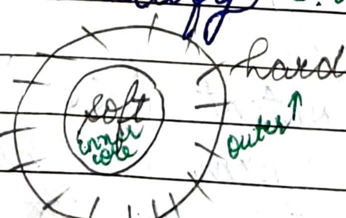
1) Extrusion Nozzles

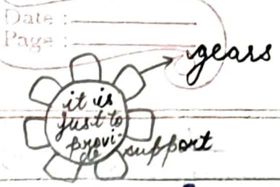
2) Bearings
eg (water tap)
Eg:- cycle chain
(overcomes friction & continue rotatary motion)

→ Grey C.I.
• C% → 3-4%.
• high Si content. है
• Slow cooling होगी
⇓ इसलिए
• soft ⇓ इसलिए ये soft होगा और
⇓ इसलिए slow cooling होगी ~~क्योंकि~~ carbon free state में present होगा।
• Presence of C → free state
↓
Graphite flakes → Grey colour
• Good vibration damping
• Self lubricating property.

* Applications.
Eg:- Springs are used (Governers.) to reduce vibration damping
1) Lathe machine bed Eg:- hydraulic structure (door closer)
2) Gear box
3) Cylinders ←
4)

3) Chilled C.I. :-
• It is obtained from gray cast iron.
• Fast cooling of Gray Cast Iron.
↓ करके उसको
Mould में डालते हैं
↓ और फिर उसको
Solidify ठंडा होने देते हैं





* Application

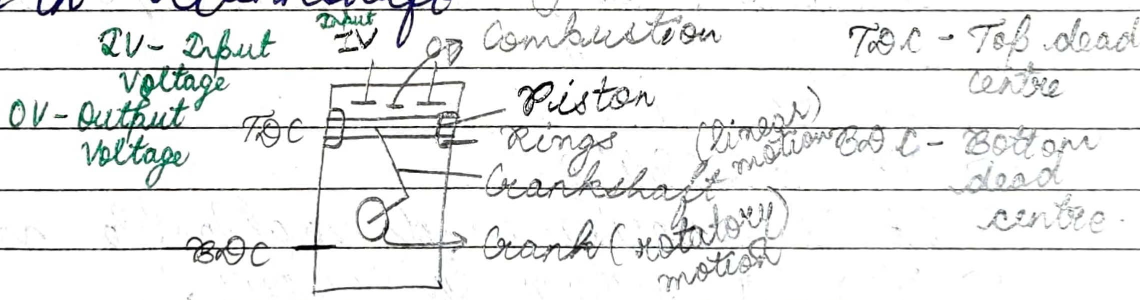
Used in gears (to change or accelerate the speed)

4) Ductile C.I.

- By adding the alloying elements like Li, Na, K, Ba (Berium), Mg, Berium.
- Graphite flakes → Spheres / Nodules
- High tensile strength and Elongation properties.
- It is soft and ductile.
- High carbon content.
- %C → 3-4%.

* Application

1) Used in utensile.
2) Used in Piston Rings } part of engine
3) Used in Crankshaft



5) Malleable C.I.

- It is obtained from white C.I.
- It (white C.I.) is heated at 950°C for 1-2 days.
- It is soft & tough.
- Malleable and less ductile.
- Carbon is in the form of graphite flakes.
- It is used for less weight casting.

Date :
Page :

* Applications :-

1. Sanitary fittings
2. Metal columns
3. Gates
4. Rail lines, etc.

→ Difference b/w :-

Steel	C.I
Ductile	Brittle
high tensile strength	high compressive strength
soft	hard

Non-Ferrous Metals
less cost (highly ductile and Malleable)

→ Al, Cu, Zn, Pt, Ti, Ni, Au, Ag, Brass, Bronze. (home appliance, they are utilised in a wide way)

1. Aluminium (Al) → Iron is not present and no magnetic properties

→ M.P. :- 660°C
→ It is light in colour. → They have ~~no~~ less rusting and ~~no~~ less corrosion.
→ Very light in weight
→ Polished mirror like appearance.
→ It is used for domestic cooking utensils, tools, windows frames, ladders, bicycles, drink bottles, etc.

2. Copper (Cu) :-

→ M.P. :- 1084°C
→ It is ductile and malleable material.
→ Red or brown in colour.
→ Good conductor of heat and electricity.
→ It can be bend easily
→ copper alloys are very useful It is used for making tubes, refrigerator tubes, electric wires, electric components, utensils, at the back

3. Zinc (Zn) :-

→ M.P. :→ 419°C

Date :
Page :

→ It is having high resistance to corrosion from moisture.
→ It is a very weak metal and mainly used for coating of steel.
→ It is used as coating or screws, buckets, etc.
→ Also used to galvanize the iron based metals.

4) Tin (Sn):-
→ M.P. :→ ~~230°c~~ 232°c
→ It is very ductile and malleable.
→ High resistant to corrosion.
→ Bright Silver in appearance.
→ It is used as coating on food cans, beverages cans, foil papers and soldering purpose.

5) Lead (Pb):-
→ M.P. :→ 327°c
→ It is soft and malleable.
→ It is a heavy metal
→ It has a bluish white colour.
→ After being freshly cut, it is soon changes to dull greyish colour when exposed to air.
→ It is used in batteries, roof flashing, X-ray protection and used for its weight in many ways.

Date :
Page :

6) Nickel (Ni) :-

→ M.P. :→ 1455°C

→ It is silvery white, hard, malleable, ductile, good conductor of heat and electricity.

→ It is a radioactive metal.

→ It is toxic and harmful to life.

→ It is used to make electric wires, coins, gas turbine blades, rocket engines. (alloy of nickel) (nimonic Alloys)

→ It can resist corrosion even at high temperatures.

7) Titanium (Ti) :-

→ M.P. :→ 1668°C

→ It is extremely strong.

→ It is having high tensile strength to density ratio. (less material will give high tensile str)

→ High corrosion resistance.

→ Ability to withstand moderately high temperature without creeping.

→ It is silvery white metal.

→ It is used in aircrafts as titanium alloys ~~never~~ naval ~~ships~~ shapped space crafts missiles and armour plating.

8) Silver (Ag) :-

→ M.P. :→ 961°C

→ It is soft, white, lusturous transition metal, high electrical conductivity & highest thermal conductivity.

Date :
Page :

→ It is used for making jewellery, currency coins, mirrors, trophies, utensils

9) Gold (Au):-
M.P. :→ 1337°C
→ It is dense, soft, shiny, malleable, ductile
→ It is bright yellow colour.
→ Traditionally active.
→ It is also having the high thermal & electric conductivity
used to supply current
→ Uses :→ Jewelleries, coins, trophies, conductors

10) Brass (पीतल):- alloy of copper
→ M.P. :→ 930°C
→ It is an alloy of copper and zinc
($\underbrace{Cu + Zn}_{\text{mainly}}$ + $\underbrace{Al + Si + Mn}_{\text{in less amount}}$)
→ It is gold in colour.
→ The hardness is more than the copper & lesser than the bronze.
→ It is having the smooth & shining surface.
→ No effect of moisture.
on/off values to supply the water
→ It is utilised in utensils, values, sockets, musical components, gears, etc.

11) Bronze (कांसा):- Alloy of copper
M.P. :→ 950°C
Cu + Sn (5–25%) + Al + Zn
It is an alloy of copper & tin

→ It is metallic brown in colour.
→ It is the hardest
→ Surface is rough and dull.
→ It is used for making medals, statues and ship mechanical components.
→ It is costlier than the brass.

Metals	Decreasing Order of Melting points
1. Titanium	1668°C
2. Nickel	1455°C
3. Gold	1337°C
4. Copper	1084°C
5. Silver	961°C
6. Bronze	950°C
7. Brass	930°C
8. Aluminium	660°C
9. Zinc	419°C
10. Tin	213°C
11. Lead	~~213°C~~ 327°C

image

	θ_1	θ_2	θ_3	θ_4		
0cm	69	69	65	65		
2cm	64	65	61	60		
4cm	57	56	59	60		
6cm	49	50	[illegible]	[illegible]	51	52
8cm	42	43	44	43		
10cm	34	35				

x. 4.

3°

2°

1°